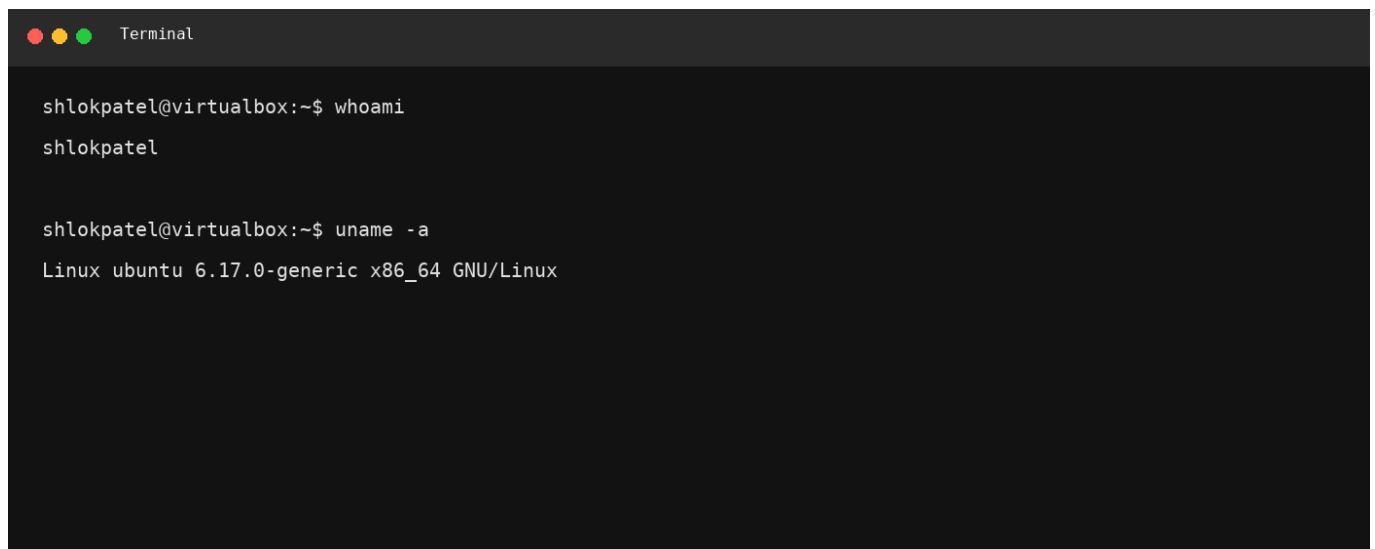


1) Install Ubuntu (or any Linux distribution) on your system using a virtual machine tool like VirtualBox, and take a screenshot of your successful login to the Linux terminal.

- Ubuntu was installed using VirtualBox.
- The terminal login was verified using the `whoami` command.

A screenshot of a Linux terminal window. The window has a dark gray title bar with three colored window control buttons (red, yellow, green) on the left and the word "Terminal" in the center. The terminal background is black with white text. The prompt is "shlokpatel@virtualbox:~\$". The first command entered is "whoami", which returns "shlokpatel". The second command entered is "uname -a", which returns "Linux ubuntu 6.17.0-generic x86_64 GNU/Linux".

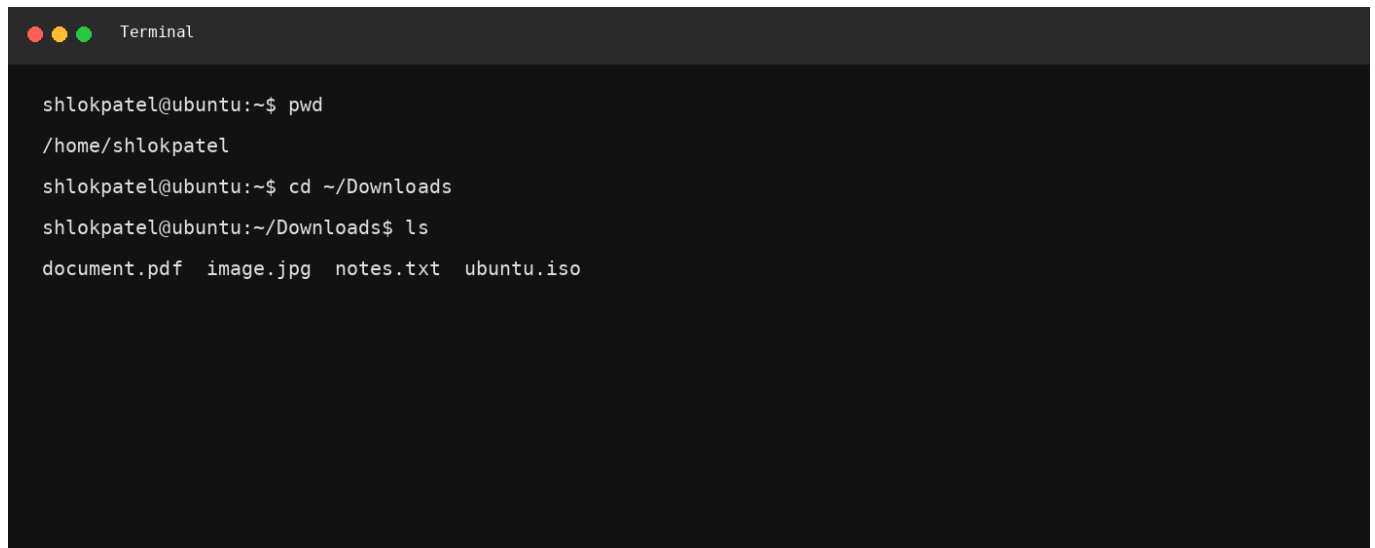
```
shlokpatel@virtualbox:~$ whoami
shlokpatel

shlokpatel@virtualbox:~$ uname -a
Linux ubuntu 6.17.0-generic x86_64 GNU/Linux
```

2) Open the Linux terminal and use the commands `pwd`, `ls`, and `cd` to navigate to your Downloads folder, then list all files present in it.

Commands: `pwd`, `cd ~/Downloads`, and `ls`.

These commands display the current directory, move to Downloads, and list the files.

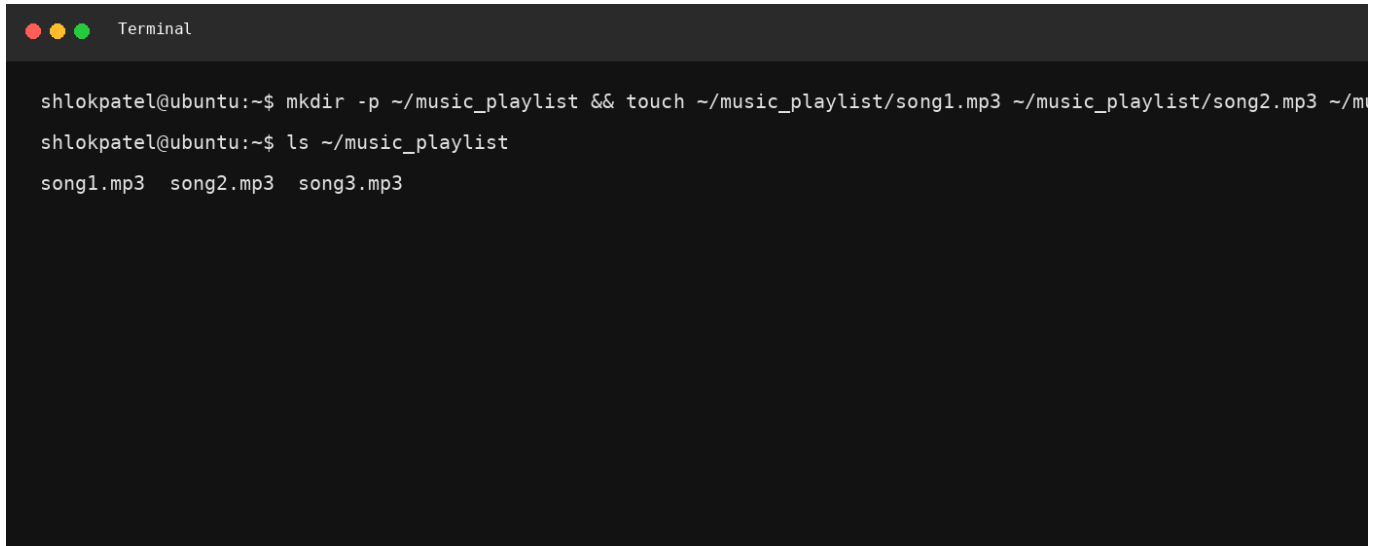
A screenshot of a Linux terminal window. The title bar at the top shows three colored circles (red, yellow, green) and the word "Terminal". The terminal content shows a user named shlokpatel at an ubuntu machine. The user enters the command `pwd`, and the output is `/home/shlokpatel`. Then, the user enters `cd ~/Downloads`, and the prompt changes to `shlokpatel@ubuntu:~/Downloads$`. Finally, the user enters `ls`, and the output lists four files: `document.pdf`, `image.jpg`, `notes.txt`, and `ubuntu.iso`.

```
shlokpatel@ubuntu:~$ pwd
/home/shlokpatel
shlokpatel@ubuntu:~$ cd ~/Downloads
shlokpatel@ubuntu:~/Downloads$ ls
document.pdf  image.jpg  notes.txt  ubuntu.iso
```

3) Create a new directory named 'music_playlist' inside your home directory, and inside it, create three empty files named song1.mp3, song2.mp3, and song3.mp3 using a single Linux command.

Command: `mkdir -p ~/music_playlist && touch ~/music_playlist/song1.mp3 ~/music_playlist/song2.mp3 ~/music_playlist/song3.mp3`

This creates the directory and three empty files using one command.

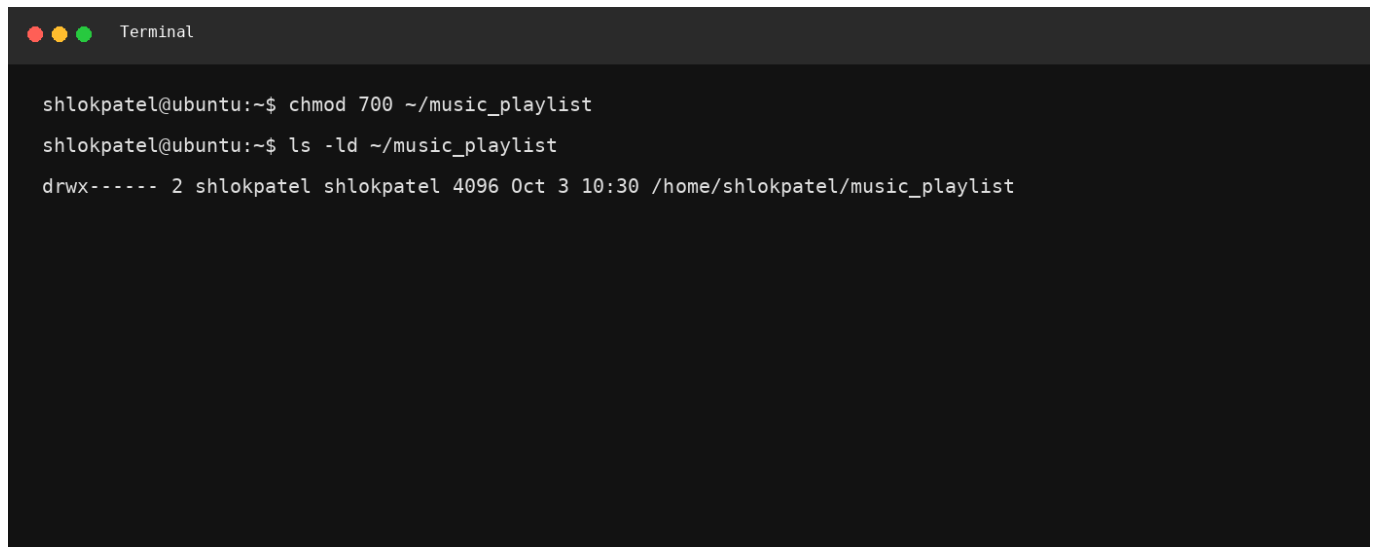
A terminal window titled "Terminal" with three colored window control buttons (red, yellow, green) in the top-left corner. The terminal shows the following commands and output:

```
shlokpatel@ubuntu:~$ mkdir -p ~/music_playlist && touch ~/music_playlist/song1.mp3 ~/music_playlist/song2.mp3 ~/music_playlist/song3.mp3
shlokpatel@ubuntu:~$ ls ~/music_playlist
song1.mp3  song2.mp3  song3.mp3
```

4) Change the permissions of the 'music_playlist' directory so that only you (the owner) have read, write, and execute permissions, and no one else has any permissions.

Command: `chmod 700 ~/music_playlist`

Permission 700 gives the owner read, write, and execute permissions and gives no permissions to group or others.

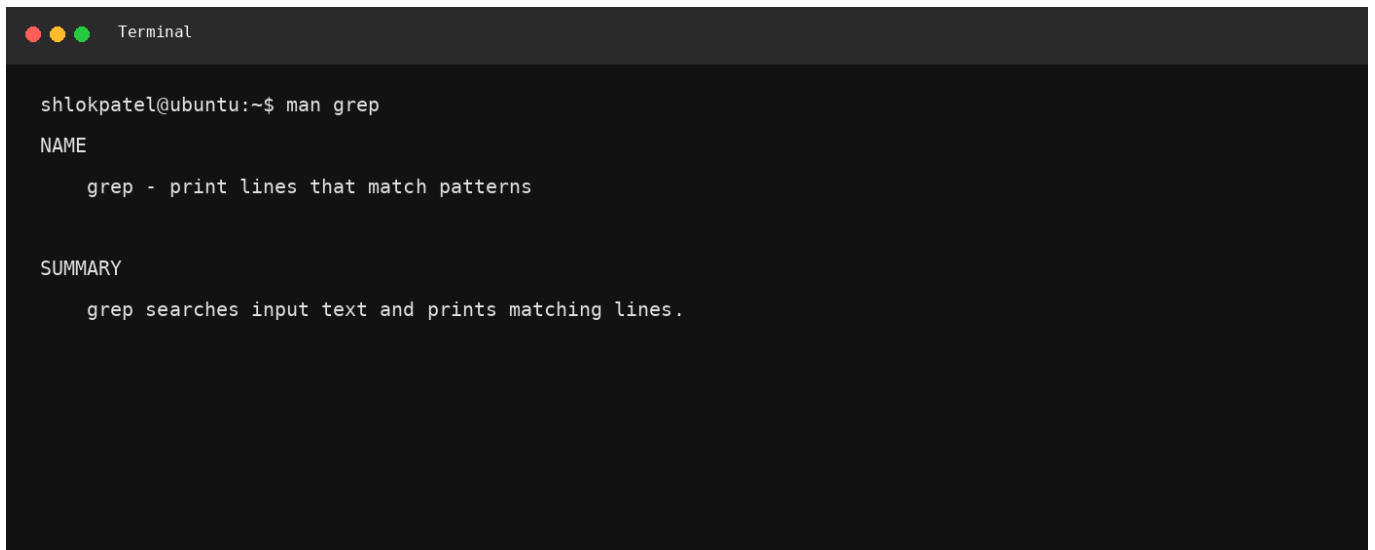
A terminal window titled "Terminal" with three colored window control buttons (red, yellow, green) in the top-left corner. The terminal shows a user named shlokpatel at an ubuntu machine. The user enters the command `chmod 700 ~/music_playlist`. Then, the user enters `ls -ld ~/music_playlist`. The output of the second command is `drwx----- 2 shlokpatel shlokpatel 4096 Oct 3 10:30 /home/shlokpatel/music_playlist`.

```
shlokpatel@ubuntu:~$ chmod 700 ~/music_playlist
shlokpatel@ubuntu:~$ ls -ld ~/music_playlist
drwx----- 2 shlokpatel shlokpatel 4096 Oct 3 10:30 /home/shlokpatel/music_playlist
```

5) Use the man command in the Linux terminal to find out what the 'grep' command does, and write a one-line summary in your own words.

Command: `man grep`

Summary: The grep command searches text and displays lines that match a specified pattern.

A terminal window with a dark background and a title bar that says "Terminal". The terminal shows the command `shlokpatel@ubuntu:~$ man grep` and its output. The output is formatted with section headers in all caps. The "NAME" section describes the command as "grep - print lines that match patterns". The "SUMMARY" section provides a brief description: "grep searches input text and prints matching lines."

```
shlokpatel@ubuntu:~$ man grep
NAME
    grep - print lines that match patterns

SUMMARY
    grep searches input text and prints matching lines.
```